Shubhadeep Chakraborty

Physics PhD Candidate, ICTS–TIFR · Stochastic Processes, Scientific Computing & AI Evaluation

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SUMMARY

Physics PhD candidate at ICTS–TIFR working in non-equilibrium statistical physics: stochastic processes, exact and asymptotic analysis, and large-scale Python simulation on HPC clusters. First-author article in the *Journal of Statistical Physics* and co-author of TheoPhysEval, a benchmark for AI agents in theoretical physics. Part-time contract experience in LLM evaluation: red teaming, RLHF feedback and supervised fine-tuning data.

SKILLS

Mathematics	Probability and stochastic processes; random walks and extreme-value statistics; asymptotic analysis; linear algebra and spectral methods; ODEs and PDEs; statistical mechanics.
Computation	Monte Carlo sampling; molecular-dynamics and event-driven simulation; Runge–Kutta integration; exact diagonalisation; numerical quadrature and root finding; averaging over thousands of realisations.
Tools	Python (NumPy, SciPy, Matplotlib, Jupyter), C, Bash, Git/GitHub, Linux, HPC batch jobs, LaTeX, Qiskit.

EXPERIENCE

Mercor — AI Safety Red Teaming, Assamese Language Expert Aug 2026 – Present

Contract · Remote

- Tested model safety in Assamese with prompt injection, persona hijacking, fictional framing, encoding and obfuscation, multi-turn escalation and iterative adversarial probing.
- Evaluated model responses for reinforcement learning from human feedback (RLHF) and rewrote them into target answers for supervised fine-tuning (SFT).

Turing — Physics Benchmark Author & Evaluator Jul – Aug 2026

Contract · Remote

- Authored **6 accepted** in-context learning tasks testing model reasoning in classical mechanics, statistical physics and quantum systems.
- Wrote reference solutions and grading criteria; analysed model errors and revised tasks for correctness, solvability, clarity and difficulty.

micro1 — STEM Domain Expert May – Jun 2026

AI training-data contract · Remote

- Annotated STEM content for multimodal AI training data and reviewed technical artefacts for relevance, quality and data-safety requirements.

RESEARCH

Hydrodynamics of the Fermi–Pasta–Ulam–Tsingou Chain 2026 – Present

ICTS–TIFR, with A. Dhar · Published in Journal of Statistical Physics

- Derived Euler and Navier–Stokes–Fourier equations for nonlinear lattices and verified Riemann-problem solutions (shocks and rarefactions) against molecular dynamics.
- Built NumPy/SciPy solvers for Gibbs-state Monte Carlo sampling, RK4 integration and equations of state; ran parameterised HPC jobs averaged over **thousands of realisations**. Code: github.com/shubhadeep763/fput_hydrodynamics.
- Current focus: ballistic spikes in the isobaric Riemann problem.

Jamming of Persistent Active Particles in One Dimension 2025 – Present

ICTS–TIFR, with A. Dhar, A. Kundu and C. Dasgupta

- Obtained exact, closed-form steady-state cluster statistics by mapping jammed configurations onto constrained lattice paths and using reflection-principle methods.
- Mapped the mean jamming time onto the minimum of a random-walk bridge; an asymptotic analysis of this minimum gives its scaling with system parameters.
- Confirmed all predictions with event-driven molecular-dynamics simulations in Python. Talk: youtu.be/VsngODq543s.

TheoPhysEval — Co-author & Problem Contributor 2026

Research collaboration · arXiv:2610.16733

- Contributed problems to a benchmark of **50 research-level** theoretical physics problems with verifiable answers, used to evaluate AI agents with literature-search and coding tools.

Fluctuations in Single-File Diffusion 2025

- Computed current and tracer cumulants for one-dimensional diffusive systems with macroscopic fluctuation theory, relating Brownian particles, hard rods and exclusion processes.

Quantum Walks & Spin Transport 2024 – 2025

- Analysed quantum query complexity for NAND-tree evaluation and simulated quantum walks on welded-tree and hierarchical graphs in Qiskit (project grade A+).
- Computed Heisenberg-chain response functions analytically and by exact diagonalisation; studied spin transport with generalized hydrodynamics.

PUBLICATIONS & PREPRINTS

- **S. Chakraborty** and A. Dhar. "Hydrodynamics of the Fermi–Pasta–Ulam–Tsingou chain." *Journal of Statistical Physics* **193**, 128 (2026). doi:10.1007/s10955-026-03690-3 · arXiv:2606.04674.
- A. Ramesh Chandra, A. Mahajan, A. Sanyal, M. Wang, **S. Chakraborty**, et al. "TheoPhysEval: Assessing AI agents at autonomous theoretical physics problem solving." *arXiv preprint* (2026). arXiv:2610.16733.

EDUCATION

International Centre for Theoretical Sciences, TIFR, Bengaluru Aug 2023 – Present
Integrated M.Sc.–Ph.D., Physics. M.Sc. coursework completed; **CGPA 8.1/10**, 102 credits. Advisor: Prof. Abhishek Dhar.
Selected coursework: Quantum Algorithms and Complexity Theory (A+); Mathematical Methods (A+); Advanced Classical Mechanics (A+); Numerical Methods (A); Advanced Statistical Mechanics; Many-Body Physics I & II.

St. Xavier's College (Autonomous), Kolkata 2020 – 2023
B.Sc., Physics. INSPIRE Scholar, Department of Science and Technology, Government of India.

TEACHING & PRESENTATIONS

- **Teaching Assistant**, Bangalore School of Statistical Physics, ICTS–TIFR (Jul 2025): led graduate tutorials in statistical mechanics, stochastic processes, hydrodynamics and transport.
- **Teaching Assistant**, Advanced Classical Mechanics, graduate course, ICTS–TIFR (ongoing).
- **Talk**, ICTS in-house meeting: "Active Jamming Problem" (2026). **Poster**, Indian Statistical Physics Community Meeting (ISPCM 2026).

HONOURS & LANGUAGES

- JEST Physics 2023: All India Rank 8 · IIT JAM Physics 2023: All India Rank 9 · INSPIRE Scholarship (DST, Government of India).
- Languages: English and Bangla (native/full professional); Assamese and Hindi (advanced).